Juri Westendorf

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Personal Statement

Second-year PhD researcher applying deep learning and statistical modelling to large-scale biological data, funded by the BBSRC SoCoBio DTP. My PhD work involves building generative and predictive models over biological sequence (DNA, RNA and protein) data using state-of-the-art pre-trained foundation models, building reproducible data pipelines, and capturing biological signal, particularly levels of protein translation from high-dimensional, noisy datasets. My background includes wet-lab work in tissue processing, fluorescence microscopy, and quantitative image analysis alongside computational biology.

Key skills at a glance:
Python/PyTorch • Statistical modelling (R) • Image-based quantification • Data pipeline design
Reproducible workflows • Cross-validation and model evaluation • Git • Laboratory data collection

Education

South Coastal Biosciences Doctoral Training Partnership (SoCoBio DTP) *Sept 2024 – Present*
University of Southampton

- Built an end-to-end deep learning pipeline that uses sequence embeddings from DNA, RNA and protein pre-trained foundation models and applies supervised learning, based on RNAseq and Riboseq derived labels, to predict tissue-specific levels of protein translation, from mRNA transcript sequence
- Built reproducible computational pipelines (GitHub) that include automated web search, annotating and processing open-source experimental data and extracting embeddings from pre-trained models
- Applied dimension-reduction and feature-importance analysis (SHAP, ElasticNet) to identify most predictive features of the input data (e.g., RNA motifs)
- Developing a data visualisation and lookup tool for exploring model outputs (gene × tissue × organism)

BSc (Hons) Neuroscience with Psychology, First Class (3.22/4.00) *Sept 2019 – Jul 2023*
University of Aberdeen

- First-class performance across chemistry, physiology, and life science modules

Research Experience

1st-Year PhD Rotation, Baker Lab for Nucleic Acid Research *Feb – Jun 2025*
University of Southampton

- Developed reactive probes to investigate RNA–protein interactions through oligonucleotide synthesis, HPLC purification, and mass spectrometry validation

Honorary Research Partner, Kang-Berg Lab *Sept – Nov 2023*
University of Aberdeen

- Quantitative image analysis of tissue samples using fluorescence microscopy, with transcriptomic and proteomic data analysis
- Co-authored manuscript in preparation on maternal inflammation and cortical development

Honours Project, Kang-Berg Lab *Jan – Apr 2023*
University of Aberdeen

- Investigated IL-17A-mediated disruption of cortical development in cerebral organoids
- Used ImageJ-based quantification and statistical analysis to identify stem cell population-specific effects

Research Assistant, Platt Lab for Neurodegeneration *Jul – Sep 2022*
University of Aberdeen

- Characterised α-synuclein pathology in a Parkinson's disease mouse model
- Conducted mouse tissue collection and homogenisation, protein quantification, Western blotting, and ImageJ-based data quantification

Research Trainee, International Research Training Group (IRTG) *Jul – Sep 2021*
RWTH Aachen University / University of Pennsylvania

- Co-conducted fMRI/HD-tDCS experiments examining working memory and resting-state neural activity

Research Skills

- **ML/DL:** PyTorch (custom training loops, classifiers, ablation studies), foundation model fine-tuning, XGBoost, SHAP, and ElasticNet feature importance
- **Data pipelines:** Reproducible processing pipelines for large biological datasets; GPU-optimised processing, structured storage, and version control
- **Programming:** Python (NumPy, Pandas), R (statistical analysis and plotting), Git, Linux/Unix
- **Quantitative image analysis:** Fluorescence microscopy quantification, ImageJ, and pattern detection in biological imaging
- **Evaluation & reproducibility:** Train/validation/test splits, cross-validation, AUROC/AUPRC, Pearson/Spearman, reproducible configurations, random seeds, and logging
- **Field/lab data collection:** Oligonucleotide synthesis, HPLC purification, tissue collection and homogenisation, cryosectioning, immunostaining, fluorescence microscopy, Western blotting, and structured experimental data collection
- **Training:** Southampton Research Software Group, Software Engineering Bootcamp (5 days); Advanced Research Software Development using AI (1 day)

Extracurricular & Work Experience

- Teaching assistant, bioinformatics and computational biology courses (2024–present)
- Selected for two start-up bootcamps (Future Worlds, University of Southampton) and a pitching competition (YES competition) for AI-driven ideas
- Student representative, Industry and Impact Committee, University of Southampton
- Four months' volunteering/teaching at a school for children with learning disabilities (Dharamsala, India, 2017) and the Basma Association of English Language (Zagora, Morocco, 2021)

Interests & Languages

- Prizes in piano and Jiu-Jitsu competitions, private pilot license for gliders
- German and English (native), French (B2), Spanish (conversational), Arabic (basics)

Publications

- Nicholas, M. T., Mehta, D., Ouyang, J., Dawoud, A., Ellison, C., Westendorf, J., … & Rackham, O. (2026). *PathPinpointR: Predicting the progression of sc-RNAseq samples through reference trajectories.* bioRxiv, 2026-04.
- Assir, M. Z., Valkila, S., Westendorf, J., Berg, D. A., Kang, E. *Spatial and temporal morphometric atlas of microglia in the developing human cortex.* (In preparation.)

References

Available upon request